

Project Aspen

Should a pension fund buy a hypothetical 20,000-hectare boreal timberland tract in Alberta, and at what price?

Answer: buy below \$109.5M (\$5,477/ha). At the \$92.0M ask the tract returns 7.5% real against a 6% hurdle, with 10% of simulated paths falling short.

HYPOTHETICAL



code + write-up

IRR

7.5%

vs 6% hurdle

NPV

\$17.7M

at 6.0% real

BREAKEVEN

\$109.5M

+ 19% vs ask

ROTATION

30 yrs

Faustmann; max-MAI 63

P(MISS HURDLE)

10%

10,000 paths

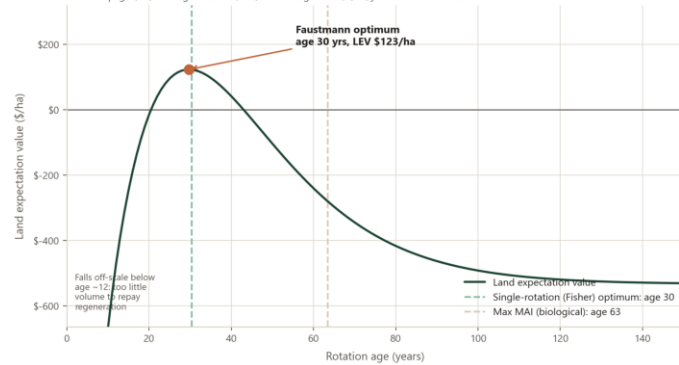
FLEXIBILITY

\$3.5M

real option on timing

Economics cut the rotation from 63 years to 30 years

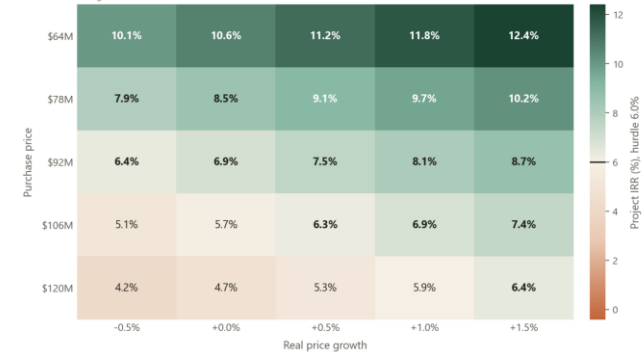
Net stumpage \$48/m³ · regeneration \$400/ha · management \$8/ha/yr · real discount rate 6.0%



Hypothetical tract, for portfolio purposes only | Illustrative price and cost assumptions

Entry price matters 3x more than the long-run price growth assumption

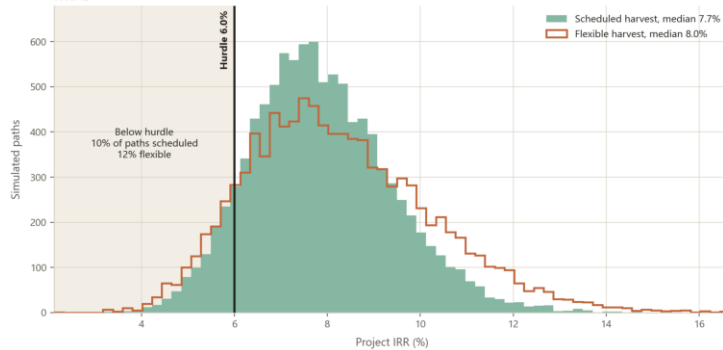
Project IRR. Moving the offer ±30% swings the IRR 5.9 points; moving real price growth across ±1 point swings it 2.3.



Hypothetical tract, for portfolio purposes only | Illustrative assumptions; every cell is a full model re-run

10% of paths miss the 6% hurdle; harvest flexibility adds \$3.5M of value

10,000 paths · mean-reverting prices, random wildfire and pest losses · scheduled P5-P95 5.6% to 10.6% · seed 42



Hypothetical tract, for portfolio purposes only | Illustrative volatility and disturbance assumptions

Timber REIT proxies move with equities: average correlation of 0.47 to the S&P/TSX Composite

Monthly total returns, Feb 1985 to Sep 2026. Listed REITs are equities first: see the proxy caveat.



Hypothetical tract, for portfolio purposes only | Real market data from Yahoo Finance

METHODS

Chapman-Richards growth curve · Faustmann rotation vs max-MAI and Fisher · 30-year DCF on an even-flow harvest schedule · 10,000-path Monte Carlo with mean-reverting prices and wildfire · Harvest-timing real option · Correlation and inflation-beta tests on real market data

TOOLS

Python · pandas · NumPy · SciPy · matplotlib · Excel (live formulas) · PowerPoint · pytest

Listed timber and farmland REITs are equities first, so the measured correlations are an upper bound and understate the diversification private timberland would deliver.

Harsit Baral · the tract is fictional; every assumption about it is illustrative and carries a source field in config/assumptions.yaml · market data from Yahoo Finance and Statistics Canada is real and cited